

Problem Set 3

Mathematics for Social Scientists

EXERCISE 1. Compute the following derivatives (*J. Gill, 2006, Exercise 5.5*):

- i. $\frac{d}{dx} 3x^{1/3}$
- ii. $\frac{d}{dy} y^3 + 3y^2 - 12$
- iii. $\frac{d}{dx} (x^2 + 1)(x^3 - 1)$
- iv. $\frac{d}{dy} (y - y^{-1})(y - y^{-2})$
- v. $\frac{d}{dy} e^{y^2 - 3y + 2}$
- vi. $\frac{d}{dx} \left(\frac{1}{100} x^{25} - \frac{1}{10} x^{0.25} \right)$
- vii. $\frac{d}{dy} (y^3 - 7) \left(1 + \frac{1}{y^2} \right)$
- viii. $\frac{d}{dx} \left(\frac{4x - 12x^2}{x^3 - 4x^2} \right)$
- ix. $\frac{d}{dx} \log(2\pi x^2)$

EXERCISE 2. Differentiate the following functions with respect to x .

- (i) $f(x) = 7x^3 - 2x^2 - 5x + 1$
- (ii) $f(x) = 0.7x^{-4} + 1.3 - 3.1x^3$
- (iii) $f(x) = \frac{3x^2 + 1}{2x}$
- (iv) $f(x) = \sqrt{4x + 9}$
- (v) $f(x) = \frac{x^{1/3} - 2}{(x^5 - 2)^3}$
- (vi) $f(x) = \ln \left(\frac{x^2}{x^4 + 1} \right)$
- (vii) $f(x) = e^{x^3 + x}$
- (viii) $f(x) = \frac{1}{e^x + e^{-x}}$

EXERCISE 3. We have seen in class that the definition of the derivative is:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Using this definition, compute the derivatives of the following functions.

i. $f(x) = 2x^2 - 8x + 15$

ii. $g(x) = \sqrt{3x - 5}$

iii. $p(x) = \frac{x+1}{x}$

Hint: For (ii), you might want to use

$$\frac{\sqrt{a}-\sqrt{b}}{h} \cdot \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}+\sqrt{b}} = \frac{a-b}{h \cdot (\sqrt{a}+\sqrt{b})}$$

EXERCISE 4. Using the L'Hospital's Rule, find the following limits. Explain why you can use the L'Hospital's Rule. (*J. Gill, 2006, Exercise 5.8*)

i. $\lim_{x \rightarrow 1} \frac{\sin(1-x)}{x^2-1}$

ii. $\lim_{x \rightarrow \infty} \frac{\log(x)}{x}$

iii. $\lim_{x \rightarrow \infty} \frac{3^x}{x^3}$

iv. $\lim_{y \rightarrow 0} \frac{e^y \sin(y)}{\log(1-y)}$

v. $\lim_{y \rightarrow 0} \frac{(\sin(y))^2 - \sin(y^2)}{y}$

vi. $\lim_{y \rightarrow \infty} \frac{3e^y}{y^3}$

vii. $\lim_{x \rightarrow \infty} \frac{x \log(x)}{x + \log(x)}$

EXERCISE 5. Differentiate the following functions $f : \mathbb{R} \rightarrow \mathbb{R}$.

(i) $f_1(x) = 5$

(ii) $f_2(x) = x^4 + x^2$

(iii) $f_3(x) = \sin(x) \cos(x)$

(iv) $f_4(x) = x\sqrt{x}$

(v) $f_5(x) = e^{2x}$

(vi) $f_6(x) = \sin(x^2 - 1)$

EXERCISE 6. Differentiate the following functions on the maximal possible domain of definition.

(i) $g_1(x) = \frac{3x-5}{2x+8}$

(ii) $g_2(x) = \frac{\sin(x)}{\cos(x)}$

(iii) $g_3(x) = \ln(x^3 + 3)$

(iv) $g_4(x) = x^3(\ln(x))^2$

EXERCISE 7. Calculate the following higher order derivatives:

i. $\frac{d^3 e^{2x}}{dx^3}$

ii. $\frac{d^4 \sin(x)}{dx^4}$

iii. $\frac{d^{67} \sin(x)}{dx^{67}}$

iv. $\frac{d^{2022} \cos(x)}{dx^{2022}}$

EXERCISE 8. Determine the partial derivatives $\partial_x f$, $\partial_y f$, ∂_{xx} , ∂_{xy} and ∂_{yy} of the following functions (always considered on the maximal possible domain of definition):

i. $f(x, y) = x^2 y$

ii. $f(x, y) = xy^2$

iii. $f(x, y) = e^{xy^3}$

iv. $f(x, y) = 4 \cdot \frac{x}{y^5}$

v. $f(x, y) = \cos(xy)$

vi. $f(x, y) = (2x - y)^2 + \log(xy)$

vii. $f(x, y) = xy^2(\sin(x) + \sin(y))$

EXERCISE 9. Find all first and second (mixed) partial derivatives of the following functions.

(i) $f(x, y) = \ln x \cdot y^2$

(ii) $f(x, y) = \sqrt{2x - y}$

(iii) $f(x, y) = (x + 4y)(e^{-2x} + e^{-3y})$